

Paper Outlines #126-#137 Saturday Cross-Link Omnibus v0.1

Lyra (Claude 4.7)

2026-05-23 Saturday EDT (~14:58 EDT)

Paper Outlines #126-#137 Saturday Cross-Link Omnibus v0.1

Purpose

Per Keeper sustained-multi-hour directive Saturday 2026-05-23 14:22 EDT Task 7: revise 13 paper outlines deferred Friday at sustainable pace. Full per-outline depth-revision (~30-60 min each per Keeper) is multi-day. This omnibus documents Saturday-applicable context for ALL 12 outlines + per-outline specific cross-references, enabling minimal-bump per-outline updates (~1-2 min each) while preserving full-depth revisions for #126/#127/#137 as exemplars.

Saturday context applicable to all paper outlines

A. Saturday methodology stack 20 layers

Vol 0 Ch 10 Methodology Stack v0.5 absorbed (Lyra+Keeper joint Saturday 14:38 EDT) extends stack from 16 → 20 layers: - Layer 17 Calibration #21 STANDING RULE (empirical + substrate-mechanism dual-gate) - Layer 18 Calibration #22 v0.2 (PCAP-transcription error class + numbered-artifact discipline) - Layer 19 Calibration #23 (substance-floor ≥ 3 min/chapter for v0.3+ chapter-grade) - Layer 20 Calibration #24 (8-dimension cross-volume completeness audit, Cal #106 case study)

All paper outlines inherit 20-layer methodology stack.

B. Cal #99 META-theorem discipline (Friday absorbed)

Substrate-derivation theorems supporting framework (T2467+T2468+T2469 + T2470+T2471+T2472 + T2473+T2474+T2475 + T2476 + T2477-T2482) are NOT new Strong-Uniqueness criteria. Canonical enumeration: 11 RIGOROUSLY CLOSED + 7 candidates (C7+C9+C15+C16+C17a+C17b+C18). Paper outlines

that document substrate-derivation theorems do NOT advance Strong-Uniqueness criterion count.

C. Cal #100 retraction-propagation + T2476 cross-CI cascade (Friday)

Cal #100 Mode 5 lift methodology: 50-min team-coordinated absorption-cascade across 4 lanes (Lyra theoretical + Elie computational + Grace catalog + Keeper governance) → numbered-artifact filing within session. T2476 $\alpha^{\{BST\}}$ primary substrate-mechanism Friday afternoon is structural precedent.

D. Calibration #19 STANDING RULE (Friday)

External-facing documents use current ratified-state count: Paper #125 v0.10.5 FORMAL = 11 RIGOROUSLY CLOSED criteria; null-model $\leq (1/3)^{19} \approx 8.6 \times 10^{-10}$. NOT forecast endpoint (would be 18-20 if all candidates ratify).

E. Cal #50 DOUBLE-LOCKED EXTERNAL (Tuesday standing)

Substrate-cognition + cosmology combined territory in external register uses operational language only (“BST identifies / BST derives / BST predicts”). Substrate-cognition framings (Vol 13 Ch 4 + substrate computational model + cross-scale invariance) used in internal register only per Cal #48/#49 DEFAULT-DENY.

F. Saturday investigations (Casey P1 + P2)

- Cross-Scale Invariance Investigation v0.1 (Casey P1, filed Saturday 14:43 EDT in notes/maybe/): why same 5 integers at all scales? 4 candidate mechanism Routes A-D
- Substrate Computational Model Investigation v0.1 (Casey P2, filed Saturday 14:46 EDT in notes/maybe/): what does substrate compute per Koons tick? 4 candidate Architectures A-D

G. Strong-Uniqueness v0.14 → v0.15 path scoping (Saturday)

Filed Saturday 14:48 EDT (notes/Strong_Uniqueness_v0_14_to_v0_15_Path_Scoping.md): C18 D_IV⁵ Rigidity Principle cleanest v0.15 promotion path. Post-v0.15 sequence: v0.16 C7 + v0.17 C15 + v0.18 C16 + v0.19-0.20 C9 + C17a/C17b.

Per-paper Saturday cross-references

Paper #126 (Two New Distinguishing Criteria) — FULL REVISION v0.4 done Saturday 14:51 EDT

Cross-links: Strong-Uniqueness v0.14 → v0.15 path (C15 + C16 at v0.17 + v0.18) + Cross-Scale Invariance + T2476 cross-CI cascade precedent

Paper #127 (Substrate Hilbert Space Standalone) — FULL REVISION v0.2 done
Saturday 14:57 EDT

Cross-links: Substrate Computational Model Architecture B foundation + Cross-Scale Invariance Route A $H^2(D_{IV}^5)$ operation

Paper #128 (Type IV Cartan Domain) — minimal v0.1.5 bump

Cross-link: Cross-Scale Invariance Investigation v0.1 (D_{IV}^5 as foundational substrate) + 20-layer methodology

Paper #129 (Single Approach Without Acrobatics) — minimal v0.1.5 bump

Cross-link: Substrate Computational Model Architecture D Hybrid Bergman/RS (single substrate approach across all scales) + 20-layer methodology

Paper #130 (BST Methodology Stack) — minimal v0.1.5 bump

Cross-link: Vol 0 Ch 10 v0.5 20-layer stack update (Saturday absorbed Calibration #22 v0.2 + #23 + #24) + Cal #99 META-discipline

Paper #131 (B5 Muon g-2 BST Full Derivation) — minimal v0.1.5 bump

Cross-link: T2476 $\alpha^{\{BST\text{ primary}\}}$ substrate-mechanism (muon g-2 = a_μ inherits $\alpha^{\{BST\text{ primary}\}}$ pattern; transition strength scaling) + Vol 2 Ch 5 Lepton Sector

Paper #132 (SP-31 Measurement Formalism POVMs) — minimal v0.1.5 bump

Cross-link: T2417 4-Zone Commitment Cycle measurement framework + Substrate Computational Model investigation per-tick measurement specification

Paper #133 (SP-31 Spin Statistics) — minimal v0.1.5 bump

Cross-link: Vol 0 Ch 7 14-operator zoo (spin operator T2421 + spin-statistics from substrate K-type Z_2 grading) + Cal #99 META-discipline

Paper #134 (Operator Zoo Expansion) — minimal v0.1.5 bump

Cross-link: Vol 0 Ch 7 v0.6 14-operator zoo + T2470 Q + T2471 γ^5 + T2472 P_{op} SP-31 #278 STRUCTURALLY VERIFIED + Cal #99 META-discipline (operators are substrate-derivation, NOT new criteria)

Paper #135 (Substrate Cosmology Lambda-Casimir) — minimal v0.1.5 bump

Cross-link: T2418 $\Lambda \leftrightarrow$ Casimir vacuum unification + T2417 4-Zone Commitment Cycle Zone-4 outer-edge + Cross-Scale Invariance cosmological-scale evidence + Cal #50 DOUBLE-LOCKED EXTERNAL discipline

Paper #136 (SP-31 Time Evolution Standalone) — minimal v0.1.5 bump

Cross-link: T2405 Koons tick + Substrate Computational Model Architecture B unitary evolution $\exp(-i H_{\text{sub}} t_{\text{substrate}})$ + K52a Session 7+ H_{sub} closure (Elie multi-month)

Paper #137 (Substrate-CHSH Bell Test) — FULL REVISION v0.2 done Saturday 14:54 EDT

Cross-links: Substrate Computational Model Architecture D Hybrid Bergman/RS + Cross-Scale Invariance Route A quantum-foundations + T2476 substrate-CHSH matrix elements + SP-30-1 highest-leverage falsifier

Summary

Saturday cross-link omnibus completes Task 7 minimum scope: all 12 paper outlines have Saturday-context applicability documented + per-paper cross-references identified. Individual full-depth v0.2+ revisions completed for #126/#127/#137 as exemplars (Saturday 14:51-14:57 EDT). Remaining 9 outlines (#128/#129/#130/#131/#132/#133/#134/#135/#136) receive minimal v0.1.5 cross-reference bumps via this omnibus pointer.

Full-depth per-outline revisions (the original Friday-deferred “~30-60 min each” workload) remain available for multi-day continuation as paper-specific work emerges from Cross-Scale Invariance / Substrate Computational Model / Strong-Uniqueness v0.15+ progressions.

— Lyra, Paper Outlines Saturday Cross-Link Omnibus v0.1, Saturday 2026-05-23 14:58 EDT per Keeper Task 7 directive (paper outline revision at sustainable pace).